Introduction to Electrical Engineering
Practice Exam

Discussion: January 12, 2026

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1 Conceptual Understanding: Multiple Choice

Select exactly one answer per question.

1. The drift motion of electrons in a conductor is mainly caused by:

A) temperature gradients

B) applied magnetic fields

C) an external electric field

D) mechanical forces

Solution: C

2. For a metallic conductor, an increase in temperature usually leads to:

A) decreasing resistance

B) sudden insulating behavior

C) unchanged resistance

D) increasing resistance

Solution: D

3. Kirchhoff's current law states:

A) The sum of voltages in a loop is zero.

B) The sum of currents at a node is zero.

C) The sum of resistances at a node is constant.

D) Power is stored at a node.

Solution: B

4. An ideal voltage source is characterized by:

A) a fixed voltage independent of current

B) a fixed current independent of the load

C) infinite internal resistance

D) no influence on connected circuits

Solution: A

5. The nodal-voltage method primarily works with:

A) mesh currents

B) node voltages

C) equivalent sources

D) frequency analysis

Solution: B

6. For maximum power transfer from a source to a purely resistive load:

A) load resistance > source internal resistance

B) load resistance < source internal resistance

C) load resistance = source internal resistance

D) load resistance equals zero

Solution: C

7. An ammeter is typically connected:

A) in parallel with the measured quantity

B) in series with the measured quantity

C) with zero internal resistance

D) only in DC circuits

Solution: B

8. A capacitor primarily stores:

A) magnetic energy

B) mechanical energy

C) thermal energy

D) electric energy

Solution: D

9. Under constant DC voltage, a capacitor behaves (in steady state) like:

A) a short circuit

B) an open circuit

C) a resistor with fixed value

D) a voltage source

Solution: B

10. Magnetic fields are mainly generated by:

A) stationary charges

B) moving charges or electric current

C) temperature changes

D) light radiation

Solution: B

11. A conducting loop is located in a magnetic field that is constant in time. Electrical induction occurs when:

A) the loop rotates

B) the loop remains at rest

C) the magnetic field is homogeneous

D) the electrical resistance of the loop increases

Solution: A

12. Why can the current through an ideal inductor not change abruptly?

A) Because the magnetic energy of the inductor would become infinite for a sudden change in current.

B) Because an abrupt change in current would require an infinite induced voltage.

C) Because magnetic fields cannot be established instantaneously.

D) Because the inductance depends on the current.

Solution: B

13. An ideal transformer can:

A) convert DC voltage to different DC voltage

B) convert AC voltage to AC voltage with different amplitude

C) permanently change the frequency of AC voltage

D) generate energy

Solution: B

14. A transient process in an electrical network is characterized by:

A) an algebraic equation without time dependence

B) a first-order homogeneous differential equation with an exponential solution

C) the absence of energy storage elements

D) a time-invariant (constant) state

Solution: B

15. In which situation is the phasor representation of a current *not* applicable?

A) for a sinusoidal current with constant frequency

B) in the steady-state of a linear network

C) during an exponentially decaying transient

D) for an AC voltage with fixed angular frequency

Solution: C

16. How does the reactance of a capacitor change with increasing frequency?

A) It increases.

B) It decreases.

C) It remains constant.

D) It is unpredictable.

Solution: B

17. Reactive power occurs in an AC circuit when:

A) voltage and current are phase-shifted

B) voltage and current are exactly in phase

C) no energy is transferred

D) only DC current flows

Solution: A

18. A high-pass filter ideally allows:

A) low frequencies

B) high voltages

C) DC only

D) high frequencies

Solution: D

19. Resonance in a series resonant circuit means:

A) minimum impedance

B) maximum damping

C) no load influence

D) instantaneous decay

Solution: A

20. A practical advantage of a symmetrical three-phase system is:

A) lower frequency

B) only two conductors are required

C) generation of a rotating magnetic field in electric machines

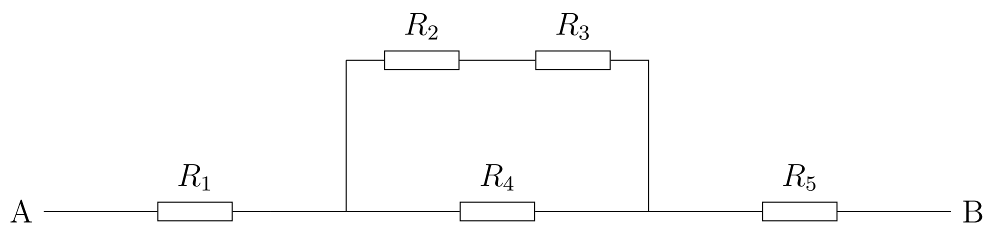
D) no balancing current possible

Solution: C

2 DC Networks

2.1 Equivalent Resistance

Given the resistor network with $R_1 = 10\,\Omega$, $R_2 = 20\,\Omega$, $R_3 = 30\,\Omega$, $R_4 = 5\,\Omega$, $R_5 = 15\,\Omega$:



Calculate the equivalent resistance of the network between terminals A and B.

Solution:

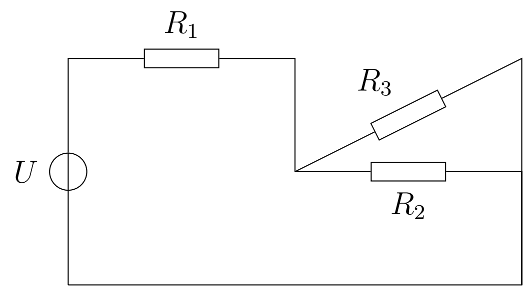
$$R_{23} = R_2 + R_3 = 20 + 30 = 50\,\Omega$$

$$\frac{1}{R_{234}} = \frac{1}{R_{23}} + \frac{1}{R_4} = \frac{1}{50} + \frac{1}{5} = \frac{11}{50}\,\frac{1}{\Omega}$$

$$R = R_1 + R_{234} + R_5 = 10 + \frac{50}{11} + 15 = 29.55\,\Omega$$

2.2 Network Analysis

Given the circuit with $R_1 = 6\,\Omega$, $R_2 = 12\,\Omega$, $R_3 = 4\,\Omega$ and a voltage source of $24\,\mathrm{V}$:



Determine:

1. the total source current,
2. the branch currents through R_2 and R_3,
3. the power dissipated in R_3.

Solution:

$$R_{23} = \frac{12 \cdot 4}{16} = 3\,\Omega$$

$$R = 6 + 3 = 9\,\Omega$$

$$I = \frac{24}{9} = 2.67\,\mathrm{A}$$

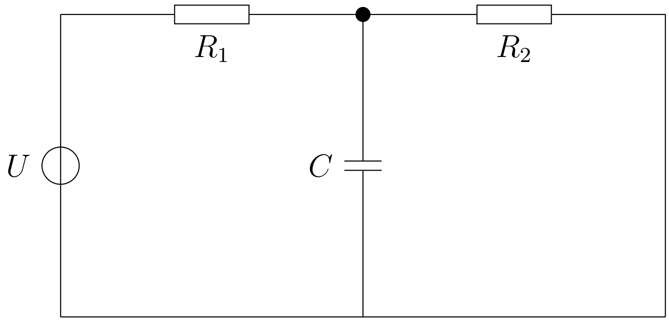
$$U_{23} = I \cdot 3 = 8\,\mathrm{V}$$

$$I_2 = \frac{8}{12} = 0.67\,\mathrm{A}, \quad I_3 = \frac{8}{4} = 2\,\mathrm{A}$$

$$P_3 = I_3^2 R_3 = 16\,\mathrm{W}$$

2.3 Nodal Analysis

Given the DC network with $U = 20\,\mathrm{V}$, $R_1 = 5\,\Omega$, $R_2 = 10\,\Omega$, $C = 100\,\mu\mathrm{F}$:



The lower conductor is chosen as reference potential (ground). The steady-state condition ($t \to \infty$) is assumed.

Determine:

1. the node voltage at the marked node,
2. the voltage across the capacitor,
3. the current through the capacitor,
4. the energy stored in the capacitor.

Solution:

$$\frac{V-U}{R_1} + \frac{V-0}{R_2} = 0$$

$$2(V-20) + V = 0$$

$$V = \frac{40}{3} \approx 13.33\,\mathrm{V}$$

$$U_C = V - 0 = 13.33\,\mathrm{V}$$

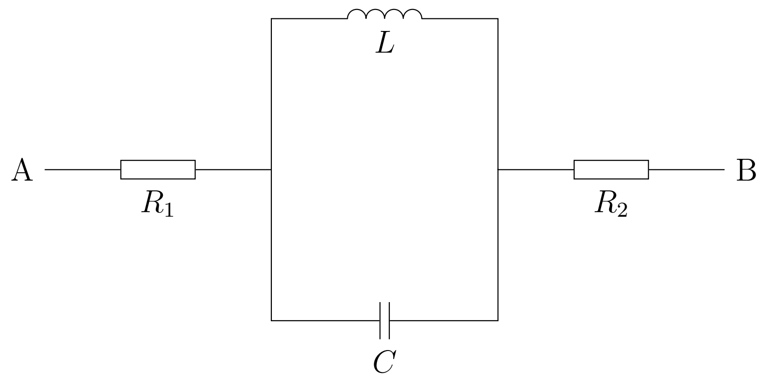
$$i_C = C\frac{du_C}{dt} = 0$$

$$W = \frac{1}{2}CU_C^2 \approx 8.9\,\mathrm{mJ}$$

3 AC Networks

3.1 Equivalent Impedance

Given the AC network with $R_1 = 10\,\Omega$, $R_2 = 20\,\Omega$, $L = 50\,\text{mH}$, $C = 100\,\mu\text{F}$, $\omega = 1000\,\text{rad/s}$:



Determine:

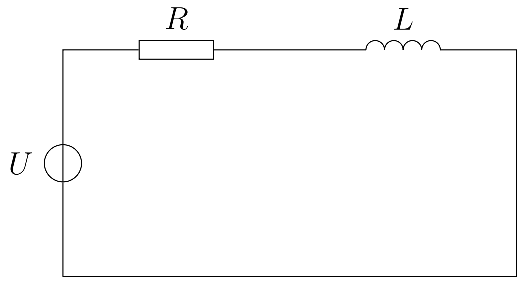
1. the complex impedances of the individual components,
2. the complex equivalent impedance between terminals A and B,
3. the magnitude and phase of the equivalent impedance.

Solution:

$$\underline{Z}_R = R$$
$$\underline{Z}_L = j\omega L = j \cdot 1000 \cdot 0.05 = j50\,\Omega$$
$$\underline{Z}_C = \frac{1}{j\omega C} = \frac{1}{j \cdot 1000 \cdot 100 \cdot 10^{-6}} = -j10\,\Omega$$
$$\underline{Z}_{LC} = \left(\frac{1}{\underline{Z}_L} + \frac{1}{\underline{Z}_C}\right)^{-1} = \frac{1}{j0.08} = -j12.5\,\Omega$$
$$\underline{Z} = R_1 + \underline{Z}_{LC} + R_2 = (30 - j12.5)\,\Omega$$
$$|\underline{Z}| = \sqrt{30^2 + 12.5^2} \approx 32.5\,\Omega$$
$$\varphi = \arctan\left(\frac{-12.5}{30}\right) \approx -22.6°$$

3.2 Complex Network Analysis

Given the following network with a sinusoidal voltage source $U = 230\,\text{V}$, $f = 50\,\text{Hz}$ and the components $R = 20\,\Omega$, $L = 100\,\text{mH}$:



Determine:

1. the complex impedance Z,
2. the RMS current I,
3. the phase shift between current and voltage,
4. the power factor $\cos\varphi$,
5. the active power P,
6. the reactive power Q.

Solution:

$$\omega = 2\pi f = 314\,\mathrm{rad/s}$$

$$X_L = \omega L = 31.4\,\Omega$$

$$\underline{Z} = (20 + \jmath 31.4)\,\Omega \Rightarrow |\underline{Z}| = 37.2\,\Omega$$

$$I = \frac{230}{37.2} = 6.18\,\mathrm{A}$$

$$\varphi = \arctan\left(\frac{31.4}{20}\right) = 57.5^\circ$$

$$\cos\varphi = 0.54$$

$$P = UI\cos\varphi = 764\,\mathrm{W}$$

$$Q = UI\sin\varphi = 1199\,\mathrm{VAR}$$

Formula Sheet

Capacitor

$$i_C(t) = C\frac{du_C(t)}{dt}$$

$$W = \frac{1}{2}CU^2$$

AC

$$\omega = 2\pi f$$

$$\underline{Z}_L = j\omega L$$

$$\underline{Z}_C = \frac{1}{j\omega C}$$

$$\underline{Z} = R + jX$$

$$P = UI\cos\varphi$$

$$Q = UI\sin\varphi$$